

Mayank Kanth

mayankkanth17@gmail.com

[Linkedin](#)

[AI PortFolio](#)

[Github](#)

[Leetcode](#)

Professional Summary

Computer Science professional with **1+ year of experience** in backend systems and AI/ML development, proficient in building scalable applications using **Java, Spring Boot, and Python-based LLM frameworks**. Experienced in working with **production-grade applications, REST APIs, and GenAI solutions**, with hands-on exposure to **OpenAI, LangChain, and AWS cloud fundamentals**. Gaining proficiency in DSA, System Design, and Real-world application development.

Work Experience

Programmer Analyst (Cognizant Technology Solutions)

Jun 2025 – May 2026

- **Optimized core business logic and RESTful endpoints** across a distributed **Spring Boot** microservices architecture, resolving **50+ critical production defects** and reducing average API response latency by **~28%**.
- **Engineered and tuned complex SQL queries and indexing** strategies across multi-million-row relational databases, accelerating query execution times by **~40%** and eliminating data inconsistency issues across downstream services.
- **Diagnosed and resolved end-to-end integration** bottlenecks across **12+ microservices** using distributed tracing and API payload validation (Postman, Splunk), slashing Mean Time to Resolution (**MTTR**) by **~35%**.
- **Collaborated with DevOps and QA in an Agile CI/CD environment** (Git, Jenkins) to deliver bi-weekly production releases and critical zero-downtime hotfixes, maintaining **99.9% system uptime** across live environments.

Software Engineer Trainee - AI/ML (Quy Technology Pvt. Ltd.)

Nov 2024 – Feb 2025

- Contributed to **POC-based AI solutions**, including a **Hotel Booking AI Assistant using LLM and vector search (AstraDB)**.
- Built **RAG-based systems using LangChain** for document retrieval and semantic search.
- Designed **FastAPI-based inference pipelines** for real-time AI applications.

Projects

Autonomous Candidate AI Copilot & Full-Stack Platform [[Project Link](#)] [[Live App](#)]

Sep 2026

- **Architected a production full-stack GenAI platform** featuring an autonomous Copilot powered by **FastAPI** and **Groq**, engineering real-time **Server-Sent Events (SSE)** streaming to achieve **<300ms time-to-first-token (TTFT)** with zero-hallucination candidate grounding.
- **Engineered production-grade security & reliability** by implementing **sliding-window conversational memory**, IP-based rate limiting, and robust prompt-injection defenses; deployed with automated CI/CD across **Railway** and **Vercel** with **99.9% uptime**.
- **Built multimodal client capabilities in React 18**, integrating hands-free **Voice STT/TTS** (Web Speech API), bilingual English/Hindi localization, and an automated **Job Description Fit Matcher** for real-time skills gap analysis.

PetPuja – AI Dining Agent [[Project Link](#)] [[Live App](#)]

Sep 2026

- Architected an autonomous dining agent (**PetPuja**) using **LangGraph**, **FastAPI**, and **Groq** (GPT-OSS-120B / 20B / Qwen-27B), separating probabilistic LLM intent classification from deterministic backend nodes (order validation, cooking, serving, and inventory mutations).
- Implemented **Human-in-the-Loop (HITL)** transactional workflows using LangGraph's **interrupt()** primitive and **MemorySaver** checkpointer, enabling pre-confirmation order modifications, proactive culinary pairing suggestions, and dynamic piece-level pricing arithmetic for multi-item portions.
- **Engineered a self-healing fault-tolerant execution** graph with per-item retry tracking, automated receipt deductions (\$0.00 charged) for partial kitchen fulfillments, and a multi-model fallback cascade to eliminate rate-limit downtime.

Tech Skills

- **Languages:** Python, Java
- **Backend:** Spring Boot, FastAPI
- **AI/ML:** OpenAI, LangChain, HuggingFace, LangGraph
- **Databases:** MySQL, Vector Databases
- **Cloud:** AWS (EC2, S3, ELB), Kafka, Redis
- **Version Control:** Git, GitHub
- **Concepts:** OOPs, DSA, REST APIs, RAG, System Design (HLD)
- **AI Assistants:** Claude Code, AntiGravity

Courses and Certifications

- **AWS Cloud Practitioner Essentials – Amazon Web Services**

Oct 2025

Education

Lingaya's Vidyapeeth (Faridabad, Haryana)

Sep 2020 - May 2024

Bachelor of Technology in Computer Science and Engineering - **CGPA : 8.47/10.0**